

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ (\text{val}^\circ \times 1) \text{snoc}$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a)

(b)

(c)

(d)

(e)

(f)

$\alpha = [\text{wrap}, \text{snoc}]$

(a) $\text{val} = ([\text{embed}, \text{op}])^\circ$ — definition
 (b) $(\varphi)^\circ = \alpha^\circ F((\varphi)^\circ) \varphi$ — catamorphisms
 (c) $(RS)^\circ = S^\circ R^\circ$ — converse
 (d) $F(R) = 1 + (R \times 1)$ — definition of F
 (e) $(R+S)[P, Q] = [RP, SQ]$ — coproduct
 (f) $[g, h]^\circ [P, Q] = g^\circ P \cup h^\circ Q$ — coproduct

(6.1b)

$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$
 op° splits a number into its quotient and remainder by 10
 $(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$
 op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
 $(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$
 embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$.
 $(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \implies \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

Theorem 6.1 (Knaster–Tarski) is (6.2b).

(6.2c)

$\alpha^\circ F(X) R \sqsubseteq X \implies (R) \sqsubseteq X$

a relation X that contains one unfolding of the fold's recursion at X contains the fold

(a)

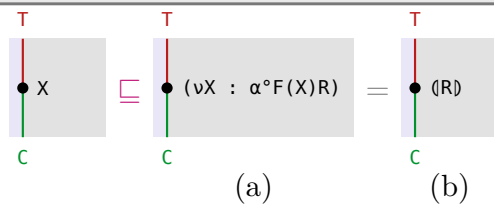
(b)

(a) (R) the least fixed point — Theorem 6.1, (6.2b)
 (b) $\alpha^\circ F(X) R \sqsubseteq X$, μ below every prefix point

$$X \sqsubseteq \alpha^\circ F(X)R \Rightarrow X \sqsubseteq \langle R \rangle$$

(6.2d)

a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

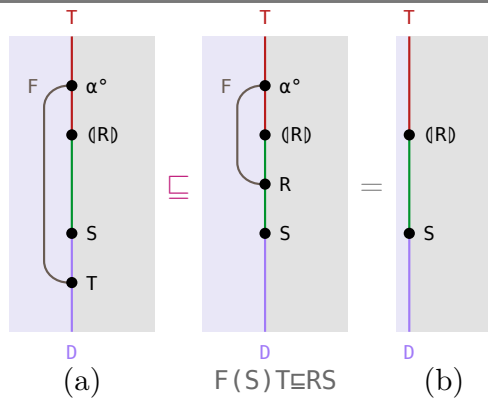


- (a) $X \sqsubseteq \alpha^\circ F(X)R$, v above every postfix point
 (b) $\langle R \rangle$ the greatest fixed point

$$F(S)T \sqsubseteq RS \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2e)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

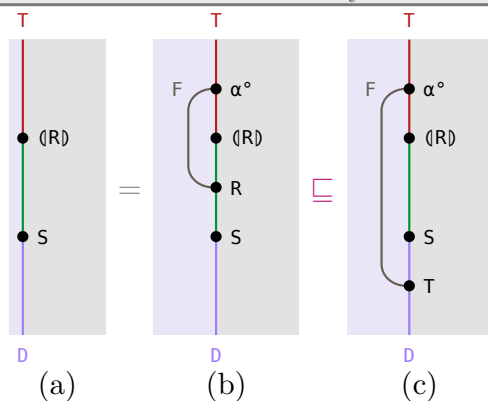


- (a) (6.2) at $X \triangleq \langle R \rangle S$
 $F(RS) = F(R)F(S) \text{ — } \triangleq \text{relator}$
 (b) $F(\langle R \rangle)R = \alpha(\langle R \rangle)$, $\alpha^\circ \alpha = 1 \text{ — } \triangleq \text{reduce}$

$$RS \sqsubseteq F(S)T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2f)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



- (a) (6.3) at $X \triangleq \langle R \rangle S$
 (b) $\alpha(\langle R \rangle) = F(\langle R \rangle)R$, $\alpha^\circ \alpha = 1 \text{ — } \triangleq \text{reduce}$
 (c) $RS \sqsubseteq F(S)T$
 $F(R)F(S) = F(RS) \text{ — } \triangleq \text{relator}$

§6.3. Hylomorphisms

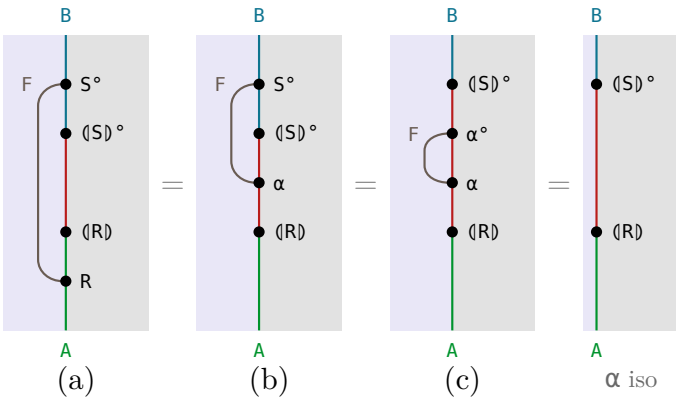
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

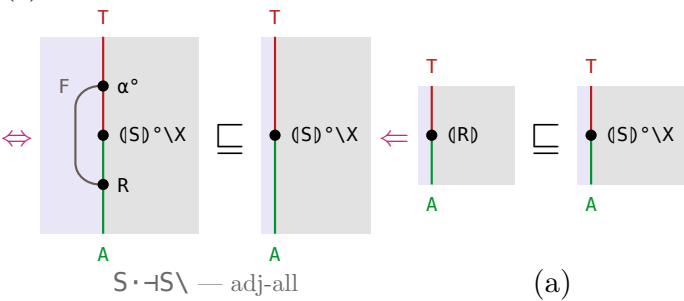
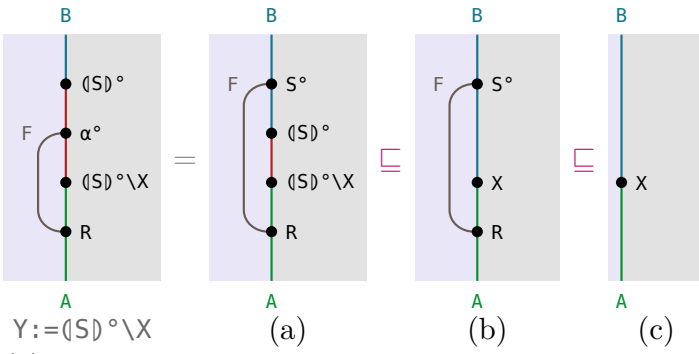
$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation

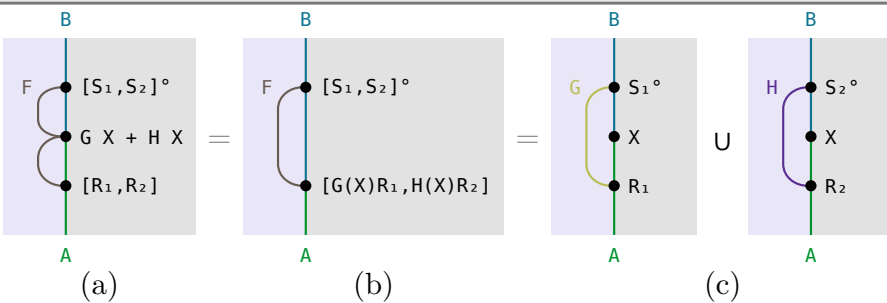


Theorem 6.2 is §Section 6.3.

$$(\forall x. \forall y. \forall x. F(X) = G(X) + H(X)) \implies ([S_1, S_2])^\circ([R_1, R_2]) = (\mu X : S_1^\circ G(X)R_1 \cup S_2^\circ H(X)R_2)$$

(6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



§6.4. Fast exponentiation and modulus computation

$\text{zero } f=g \wedge \text{shift } f=(f \times 1)h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h) \sqsubseteq f \quad (6.4a)$				
once f sends zero to g and turns shift into h , the recursion that halves the argument at each step computes f				
(a)	$\sqsubseteq$	(b)	$=$	Corollary 6.1, (6.3b)

(a) convert simple
(b) fusion: zero $f=g$, shift $f=(f \times 1)h$

$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$	
the recursion that halves the exponent at each step computes a to the power b	
zero exp(a)=one	
a to the power 0 is 1	
shift exp(a)=(exp(a)×1)op(a)	
a to the power 2n+d is op(a) of a to the power n and d	

$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$	
the recursion that halves a at each step computes the remainder of a by b	
zero mod(b)=zero	
0 mod b is 0	
shift mod(b)=(mod(b)×1)op(b)	
(2a+d) mod b is op(b) of a mod b and d	

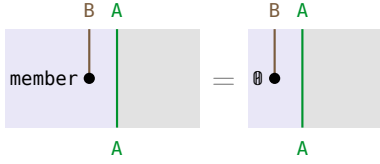
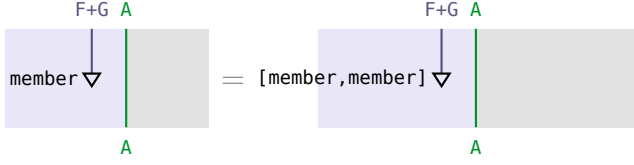
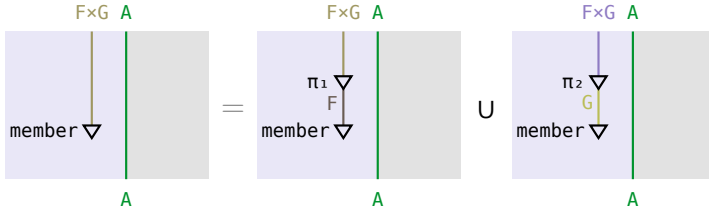
§6.5. Unique fixed points

$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R) \quad (6.5a)$				
R is inductive when $R \sqsubseteq SR$ for an inductive S				
(a)	$\sqsubseteq$	(b)	$\sqsubseteq$	(a) division by S (b) division by R

member=0

(6.5b)

a constant relator records no elements, so its membership is empty

**member=[member, member]**a member of $F+G$ is a member of whichever summand the value is in**member= π_1 member $\cup$ π_2 member**a member of $F \times G$ is a member of either component**inductive(Smember) $\wedge$ $X=SF(X)R \wedge Y=SF(Y)R \implies X=Y$**

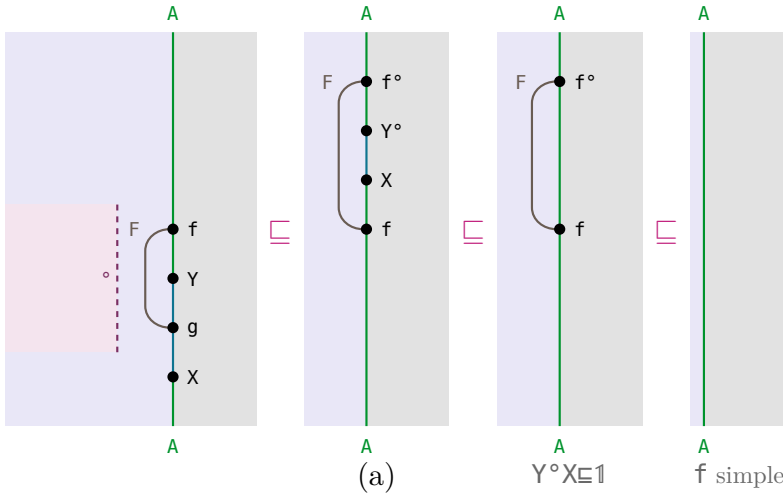
(6.5c)

when **S member** is inductive, two solutions X, Y of $X=SF(X)R$ are equal**inductive(Smember) $\wedge$ Entire(S) $\wedge$ Entire(R) $\wedge$ $SF(X)R \subseteq X \implies$ Entire(X)**

(6.5d)

when **S member** is inductive and **S, R** are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire**inductive(gmember) $\wedge$ Map(g) $\wedge$ Map(f) $\wedge$ $X=gF(X)f \implies$ Map(X)**

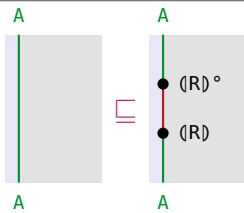
(6.5e)

when **g member** is inductive and **f, g** are maps, the solution of $X=gF(X)f$ is a map(a) $X=gF(X)f$, **g** simple

(a)

 $Y^\circ X \subseteq 1$ **f** simple**inductive(R° member) $\wedge$ $1 \in R^\circ R \implies 1 \in (R)^\circ (R)$**

(6.5f)

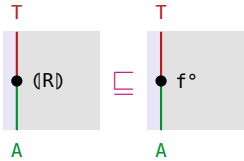
when **R° member** is inductive and **R** is surjective, the fold (R) is surjective

(a)

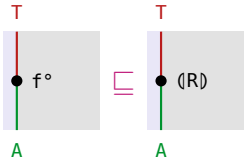
(a) (6.5d) at $S=R^\circ$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \wedge \text{inductive}(\alpha^\circ \text{member}) \Rightarrow f^\circ = \langle R \rangle \quad (6.5g)$$

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)

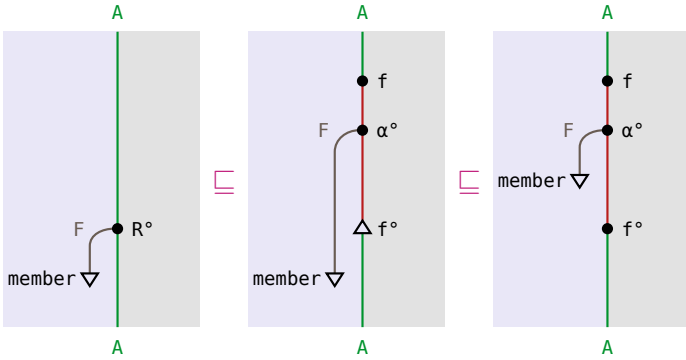
(a) shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$ 

(a)

(a) (6.5f) and $\langle R \rangle f \sqsubseteq 1$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ \quad (6.5h)$$

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)

(a) $Rf \sqsubseteq F(f)\alpha$, shunting

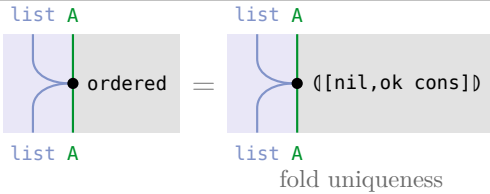
(b)

(b) member lax natural

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle \quad (6.6a)$$

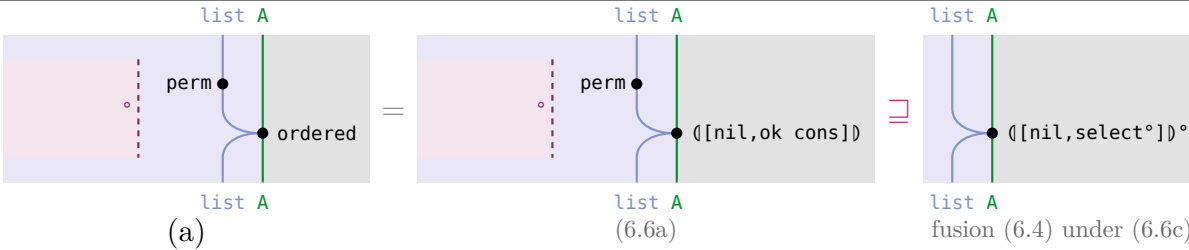
a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it



fold uniqueness

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

every output of unfolding the input by **select** is a sorted permutation of it



(a)

(6.6a)

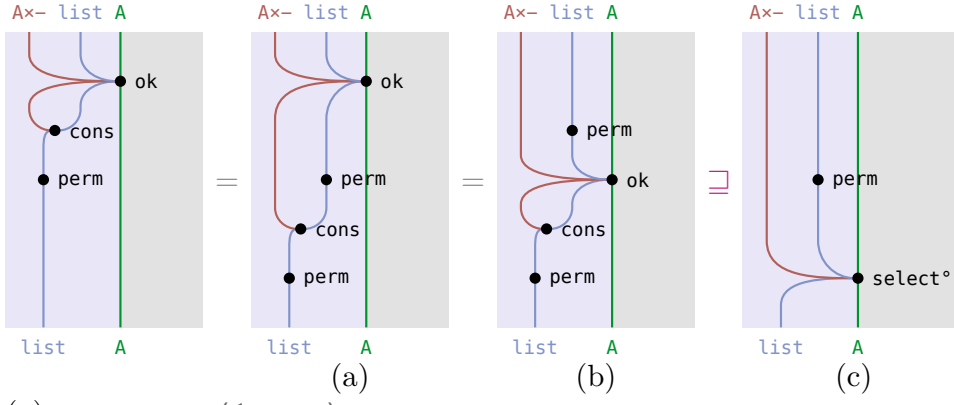
fusion (6.4) under (6.6c)

(a)

 $\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

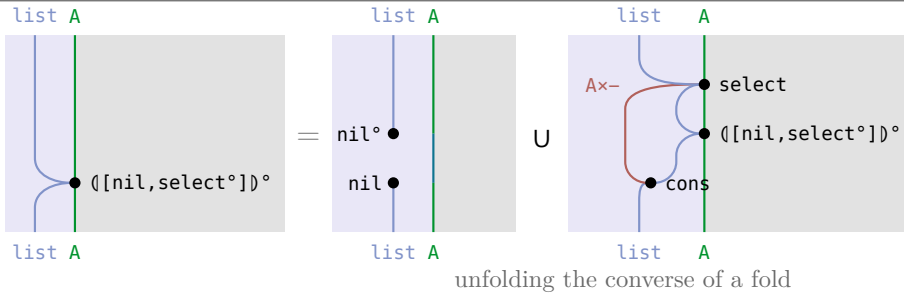
$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

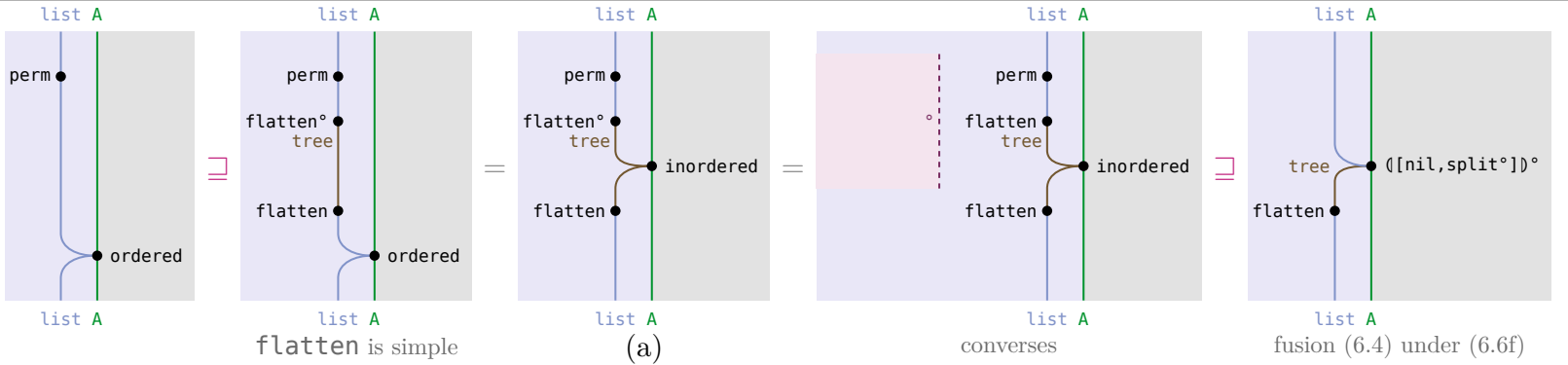
selection sort returns $[]$ on $[]$, and otherwise selects (a, y) , sorts y and conses a back on



$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

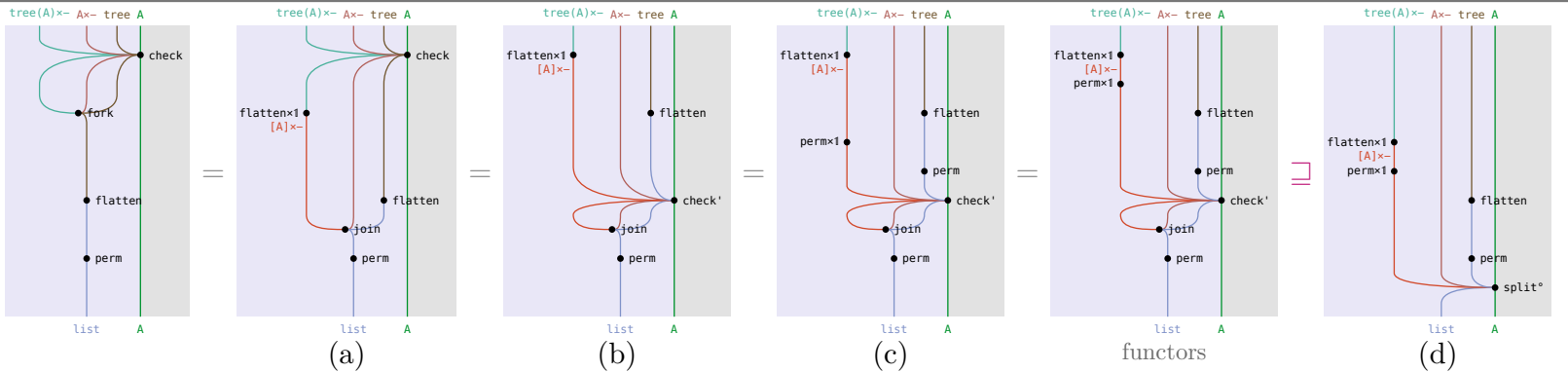
$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check}' \text{ fork flatten perm} \quad (6.6f)$$

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

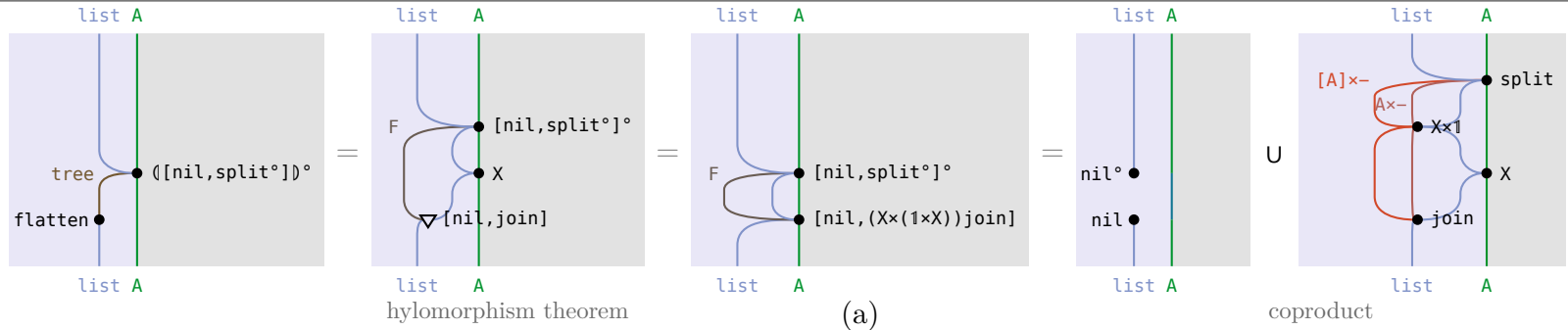


$$([nil, split^\circ])^\circ \circ flatten = nil^\circ nil \cup split$$

(6.6g)

$$((([nil, split^\circ])^\circ \circ flatten) \times (1 \times ([nil, split^\circ])^\circ \circ flatten))) \circ join$$

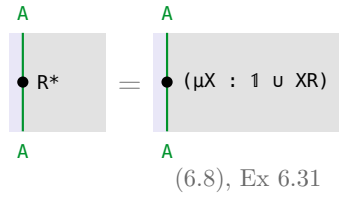
quicksort returns [] on [], and otherwise splits into (x, a, y), sorts x and y and joins them around a



§6.7. Closure

$$R^* = (\mu X : 1 \cup XR)$$

(6.7a)

closing R by composing it on the left and closing it by composing on the right give the same relation R^* 

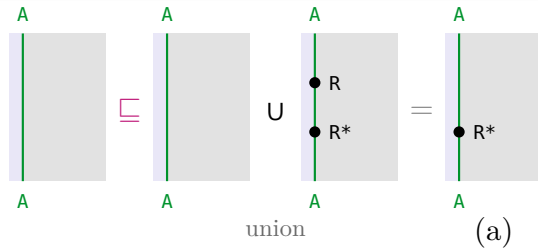
$$1 \in X \wedge XX \in X \implies R \in X \iff R^* \in X$$

(6.7b)

for a preorder X , X contains R exactly when it contains R^*

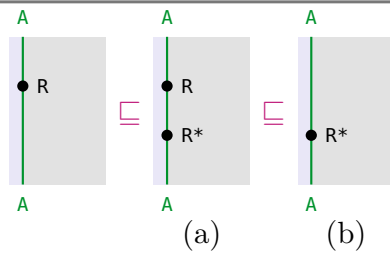
$$1 \in R^*$$

(6.7c)

 R^* is reflexive

$$R \in R^*$$

(6.7d)

 R^* contains R 

- (a) $1 \in R^*$ — (6.7c)
 (b) fixed point

(6.7e)

$R^*R^* \sqsubseteq R^*$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup RX$

(a) composition distributes over $\cup$

(b) $RR^* \sqsubseteq R^*$ — (6.7d)

(6.7f)

$1 \sqsubseteq X \wedge XX \sqsubseteq X \wedge R \sqsubseteq X \implies R^* \sqsubseteq X$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup RX$, so it contains R^*

$1 \sqsubseteq X, R \sqsubseteq X$ $XX \sqsubseteq X$

(6.7g)

$\text{inductive}(R) \wedge X = 1 \cup RX \implies X = R^*$

when R admits induction, R^* is the only X equal to $1 \cup RX$

(a) $(R^* \sqsubseteq R^*/X) \cap R^* \sqsubseteq R^*/X$

(6.7h)

$\frac{R^*}{3} = (\frac{1}{3}, \frac{RR^*}{3}) \text{ cup}$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors

(a) $\Lambda(R \cup S) = (\Lambda(R), \Lambda(S)) \text{ cup}$

(6.7i)

$R - \emptyset = R$

taking nothing away leaves R

$R \cup (S - R) = R \cup S$

beside R , only the part of S outside R adds anything

$R - (S \cup T) = R - S - T$

taking away S and then T takes away $S \cup T$

$(R \cup S) - T = (R - T) \cup (S - T)$

taking T away from a union takes it away from each part

(6.7j)

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$

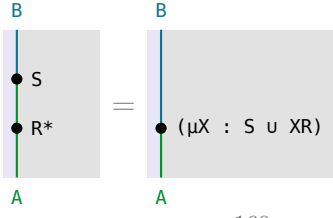
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

Ex 6.35

$$SR^* = (\mu X : S \cup XR)$$

(6.7k)

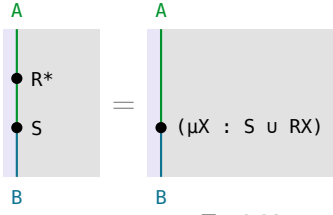
S followed by any number of R steps is the least X containing S and closed under a further R step



p.160

$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

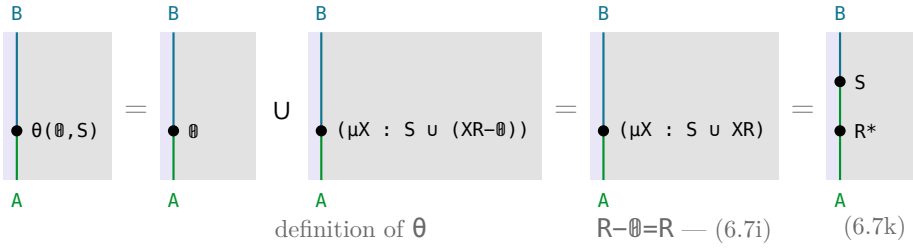


Ex 6.32

$$\theta(\emptyset, S) = SR^*$$

(6.7l)

started with nothing found and S to explore, θ computes SR^*

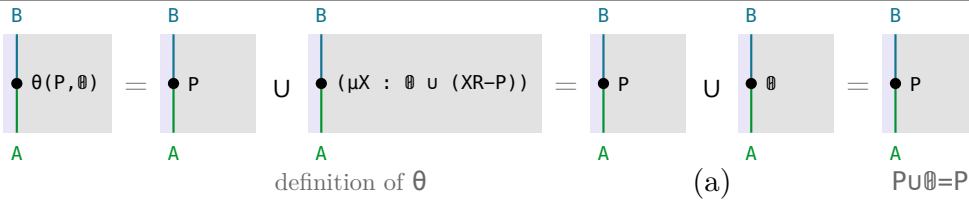
definition of θ $R - \emptyset = R$ — (6.7i)

(6.7k)

$$\theta(P, \emptyset) = P$$

(6.7m)

with nothing left to explore, θ returns what it has found

definition of θ

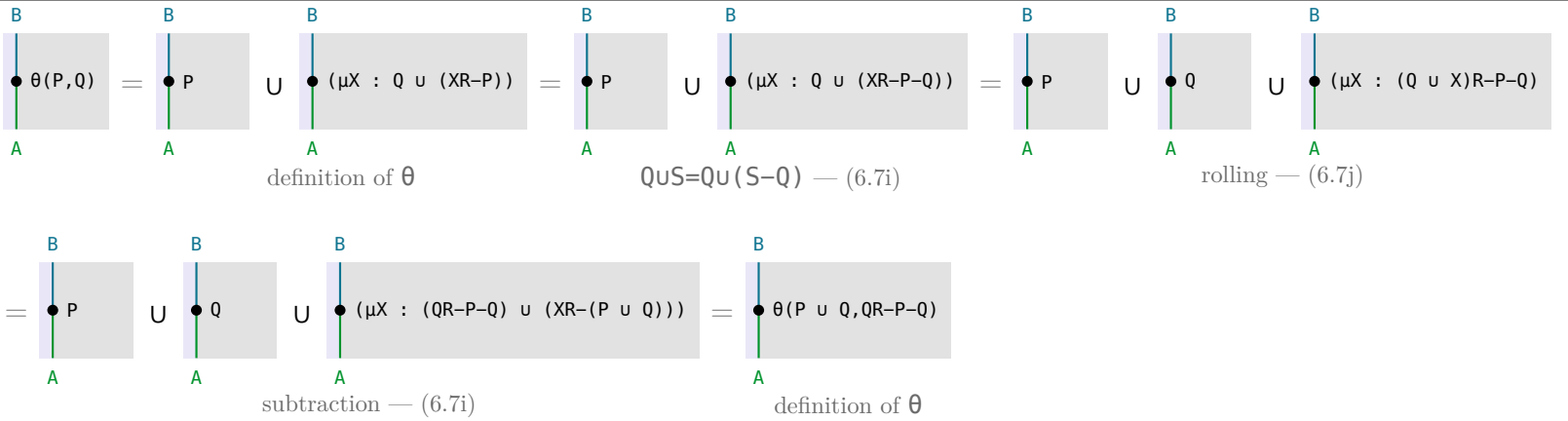
(a)

 $P \cup \emptyset = P$ (a) $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$

$$\theta(P, Q) = \theta(P \cup Q, QR - P - Q)$$

(6.7n)

one round moves Q into the found part and explores the new R -successors of Q that are in neither

definition of θ $QuS = Qu(S - Q)$ — (6.7i)

rolling — (6.7j)

subtraction — (6.7i)

definition of θ